System Architecture

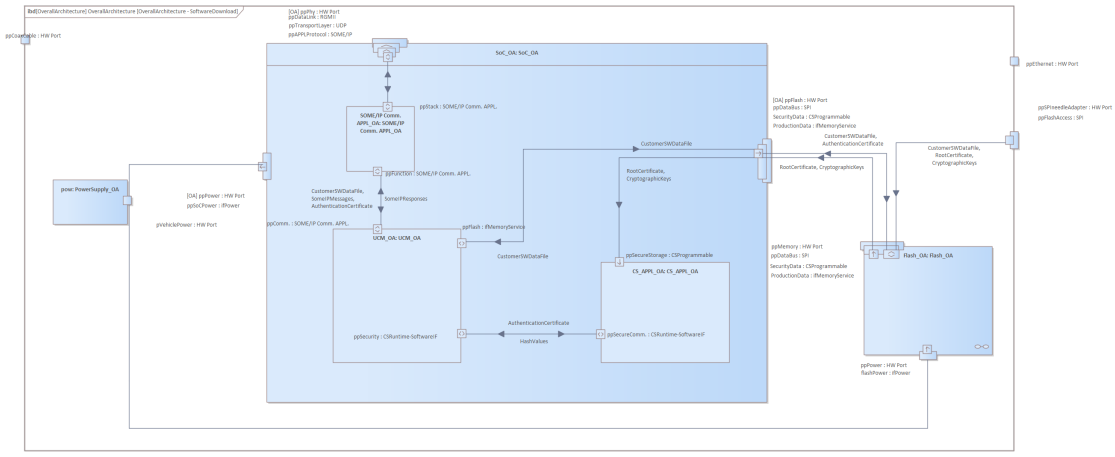
1 Overview -Overall Architecture 1
2 Heating 8
3 Shutter 14
4 Software Download 18

1 Overview -Overall Architecture

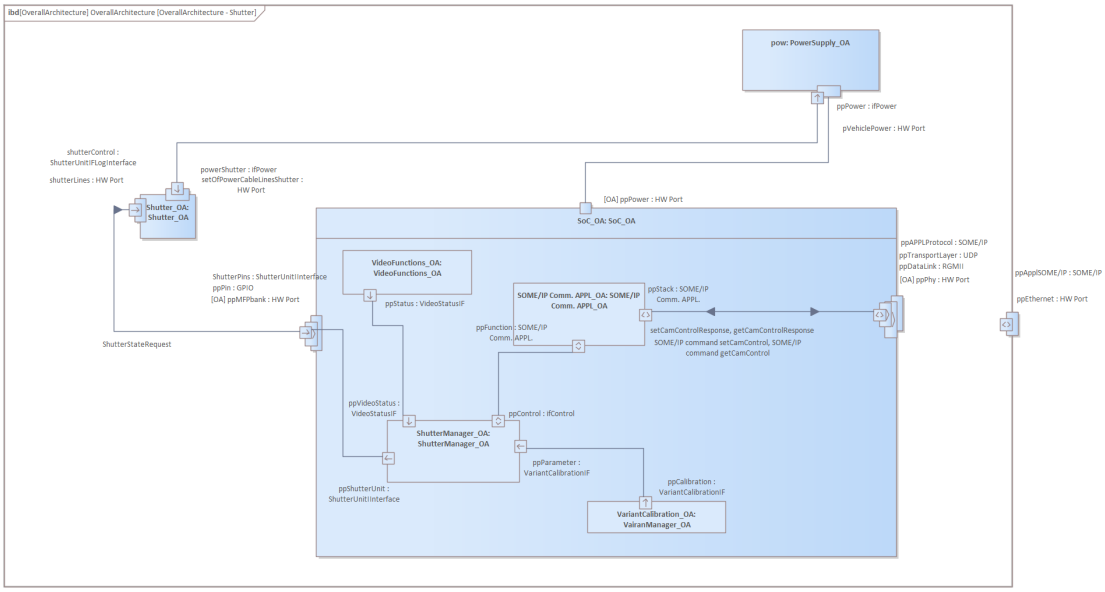
MVC-102782 - OverallArchitecture

[EA]

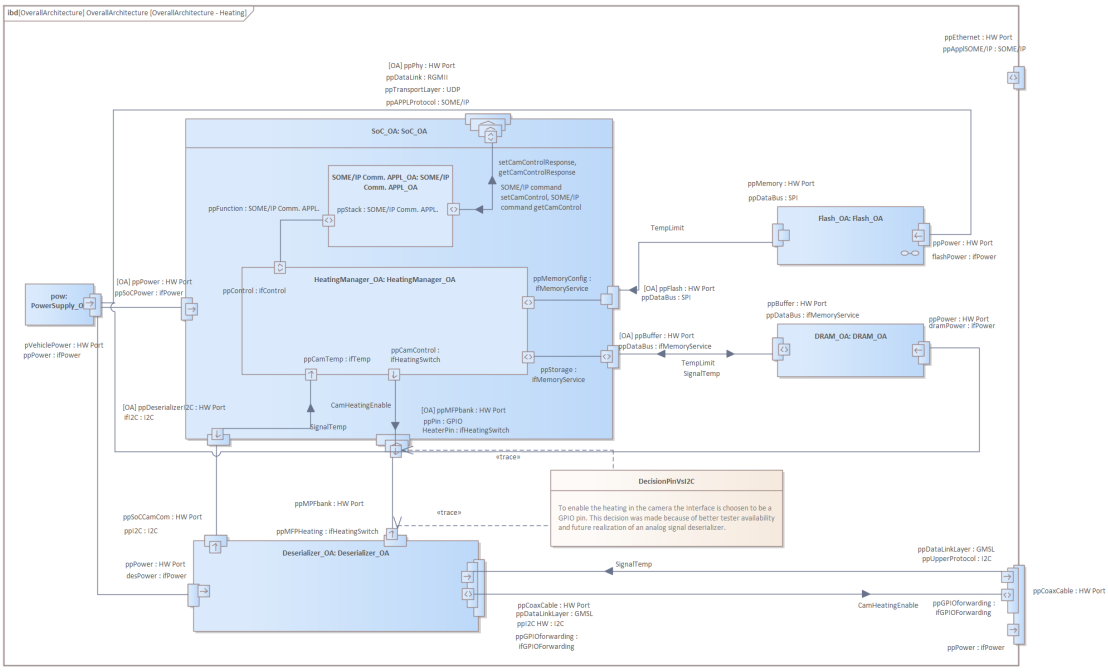
OverallArchitecture - SoftwareDownload



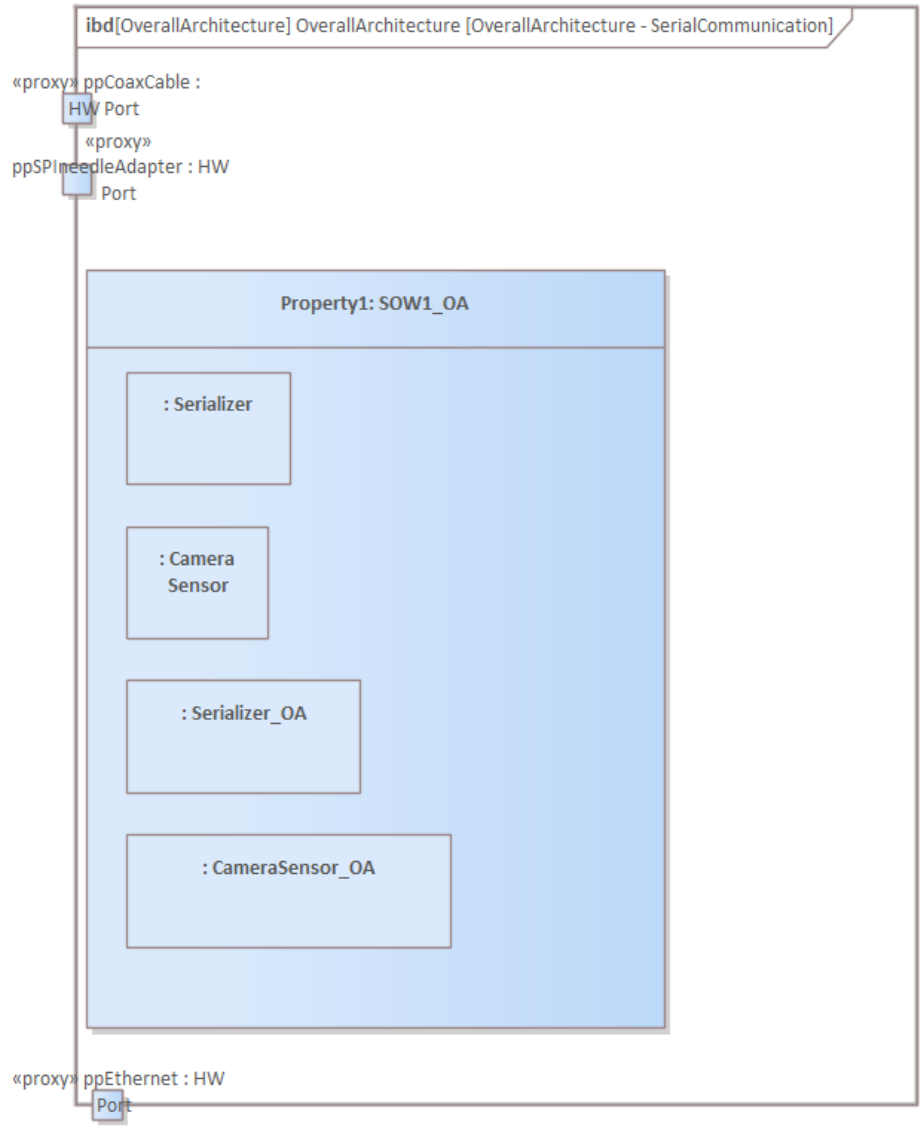
OverallArchitecture - Shutter



OverallArchitecture - Heating



OverallArchitecture - SerialCommunication



MVC-102787 - SoC_OA

[Draft **]**

MVC-105482 - VideoFunctions_OA

[Draft **]**

MVC-105483 - ppStatus

[Draft, VideoStatusIF;{5484315B-2126-44e4-9827-99711F0C7768};Type **]**

MVC-102888 - VariantCalibration_OA

[Specified **]**

MVC-105477 - ppCalibration

[Draft, VariantCalibrationIF;{D0BB15CA-31D4-4dbd-B8FB-079D82BD9218};Type **]**

MVC-105721 - UCM_OA

[Draft **]**

MVC-105722 - ppSecurity

[Draft, CSRuntime-SoftwareIF;{90712335-A2BB-CE66-8427-E3AA7E9BD2A1};Type **]**

MVC-105723 - ppFlash

[Draft, ifMemoryService;{BA0C105B-DA45-DF22-A3EF-1C30669EC9B8};Type]

MVC-105724 - ppComm.

[Draft, SOME/IP Comm. APPL.;{532DF7FD-C45A-456a-ADA7-17A757E4FEB6};Type]

MVC-105486 - SOME/IP Comm. APPL_OA

[Specified]

MVC-105487 - ppStack

[Draft, SOME/IP Comm. APPL.;{532DF7FD-C45A-456a-ADA7-17A757E4FEB6};Type]

MVC-105488 - ppFunction

[Draft, SOME/IP Comm. APPL.;{532DF7FD-C45A-456a-ADA7-17A757E4FEB6};Type]

MVC-102889 - ShutterManager_OA

[Specified]

MVC-105480 - ppVideoStatus

[Draft, VideoStatusIF;{5484315B-2126-44e4-9827-99711F0C7768};Type]

MVC-105481 - ppShutterUnit

[Draft, ShutterUnitIInterface;{B7ECECF6-6979-50E1-B638-329750D84245};Type]

MVC-105478 - ppParameter

[Draft, VariantCalibrationIF;{D0BB15CA-31D4-4dbd-B8FB-079D82BD9218};Type]

MVC-105479 - ppControl

[Draft, ifControl;{8C1DD91B-9B69-D059-8E6F-B597D2498320};Type]

MVC-105484 - ppMemory

[Draft, HW Port;{9C91EC93-E579-DAF4-84C9-608F4873EDC4};Type]

MVC-102844 - ppDataBus

[Draft, SPI;{A29539D0-FBCE-6D33-B313-4703268EED4B};Type]

MVC-105729 - SecurityData

[Draft, CSProgrammable;{8A4541D5-DECC-4146-850B-389048FDECC3};Type]

MVC-105728 - ProductionData

[Draft, ifMemoryService;{3512C79A-14DF-41b3-96AB-9EFCD06B527C};Type]

MVC-105485 - ppBuffer

[Draft, HW Port;{9C91EC93-E579-DAF4-84C9-608F4873EDC4};Type]

MVC-102896 - ppDataBus

[Draft, ifMemoryService;{BA0C105B-DA45-DF22-A3EF-1C30669EC9B8};Type]

MVC-103377 - HeatingManager_OA

[Specified]

MVC-102832 - ppStorage

[Draft, ifMemoryService;{BA0C105B-DA45-DF22-A3EF-1C30669EC9B8};Type]

MVC-102830 - ppMemoryConfig

[Draft, ifMemoryService;{BA0C105B-DA45-DF22-A3EF-1C30669EC9B8};Type]

MVC-102841 - ppControl

[Draft, ifControl;{8C1DD91B-9B69-D059-8E6F-B597D2498320};Type]

MVC-105491 - getCamControl

[Draft]

MVC-102839 - ppCamTemp

[Draft, ifTemp;{BFC63AC8-7BB7-062F-9129-2E11094548C7};Type]

MVC-102831 - ppCamControl

[Draft, ifHeatingSwitch;{D0501131-27AE-45a9-9C83-BA68889955FE};Type]

MVC-105725 - CS_APPL_OA

[Draft]

MVC-105727 - ppSecureStorage

[Draft, CSProgrammable;{850B3890-48FD-ECC3-AC56-2B6B1BBAD533};Type]

MVC-105726 - ppSecureComm.

[Draft, CSRuntime-SoftwareIF;{90712335-A2BB-CE66-8427-E3AA7E9BD2A1};Type]

MVC-103389 - [OA] ppPower

[Draft, HW Port;{9C91EC93-E579-DAF4-84C9-608F4873EDC4};Type]

MVC-103376 - ppSoCPower

[Draft, ifPower;{B4E0CCF1-C664-AC45-A778-2FF3A896CAF8};Type]

MVC-103391 - [OA] ppPhy

[Draft, HW Port;{9C91EC93-E579-DAF4-84C9-608F4873EDC4};Type]

MVC-102845 - ppDataLink

[Draft, RGMII;{BEAB1857-0B87-1E57-9D70-A7A304F950CE};Type]

MVC-102855 - ppTransportLayer

[Draft, UDP;{82196744-FE17-3977-B719-F265E874F03C};Type]

MVC-102858 - ppAPPLProtocol

[Draft, SOME/IP;{AF858678-8715-1A9D-85BB-EFA26BC923E2};Type]

MVC-102883 - [OA] ppMFPbank

[Draft, HW Port;{9C91EC93-E579-DAF4-84C9-608F4873EDC4};Type]

MVC-102827 - ppPin

[Draft, GPIO;{9261F420-9E29-755D-9475-1744C6A9E989};Type]

MVC-102854 - ShutterPins

[Draft, ShutterUnitIInterface;{B7ECECF6-6979-50E1-B638-329750D84245};Type]

MVC-102856 - HeaterPin

[Draft, ifHeatingSwitch;{D0501131-27AE-45a9-9C83-BA68889955FE};Type]

MVC-103390 - [OA] ppDeserializerI2C

[Draft, HW Port;{9C91EC93-E579-DAF4-84C9-608F4873EDC4};Type]

MVC-102895 - ifI2C

[Draft, I2C;{A5CFB26A-4616-378C-9CDD-009638957548};Type]

MVC-102790 - Shutter_OA

[Specified]

MVC-102893 - shutterLines

[Draft, HW Port;{CE73F1D4-FF0D-4864-923E-8ABB7A1FEF2F};Type]

MVC-102894 - shutterControl

[Draft, ShutterUnitIFLogInterface;{6C5EC395-08B4-4650-B7EC-ECF6697950E1};Type]

MVC-102882 - setOfPowerCableLinesShutter

[Draft, HW Port;{9C91EC93-E579-DAF4-84C9-608F4873EDC4};Type]

MVC-102849 - powerShutter

[Draft, ifPower;{B4E0CCF1-C664-AC45-A778-2FF3A896CAF8};Type]

MVC-107620 - Property1

[Draft]

MVC-107621 -

[Draft]

MVC-107622 -

[Draft]

MVC-107623 -

[Draft]

MVC-107624 -

[Draft]

MVC-105732 - ppSPIneedleAdapter

[Draft, HW Port;{9C91EC93-E579-DAF4-88C5-ECD4FF5416B9};Type]

MVC-105733 - ppFlashAccess

[Draft, SPI;{B3134703-268E-ED4B-A256-149B75FA4A20};Type]

MVC-105734 - SecurityData

[Draft, CSProgrammable;{8A4541D5-DECC-4146-850B-389048FDECC3};Type]

MVC-102785 - ppEthernet

[Draft, HW Port;{9C91EC93-E579-DAF4-84C9-608F4873EDC4};Type]

MVC-102880 - ppAppISOME/IP

[Draft, SOME/IP;{AF858678-8715-1A9D-85BB-EFA26BC923E2};Type]

MVC-102791 - ppCoaxCable

[Draft, HW Port;{9C91EC93-E579-DAF4-84C9-608F4873EDC4};Type]

MVC-102885 - ppPower

[Draft, ifPower;{B4E0CCF1-C664-AC45-A778-2FF3A896CAF8};Type]

MVC-102887 - ppDataLinkLayer

[Draft, GMSL;{A98A828F-5535-F52D-A5E7-DCD2CDC7B8F0};Type]

MVC-102864 - ppUpperProtocol

[Draft, I2C;{A5CFB26A-4616-378C-9CDD-009638957548};Type]

MVC-106230 - ppGPIOforwarding

[Draft, ifGPIOForwarding;{BD398808-97FC-70D8-94CF-CF9A361E3010};Type]

MVC-102788 - pow

[Draft]

MVC-105490 - pVehiclePower

[Draft, HW Port;{9C91EC93-E579-DAF4-84C9-608F4873EDC4};Type]

MVC-102848 - ppPower

[Draft, ifPower;{B4E0CCF1-C664-AC45-A778-2FF3A896CAF8};Type]

MVC-102789 - PHY_OA

[Draft]

MVC-103385 - ppPHYPower

[Draft, ifPower;{B4E0CCF1-C664-AC45-A778-2FF3A896CAF8};Type]

MVC-102892 - ppEthernetLines

[Draft, HW Port;{9C91EC93-E579-DAF4-84C9-608F4873EDC4};Type]

MVC-102847 - DataLink

[Draft, XYZBASE T1;{AA765D3C-A7DC-5479-A858-89EF00DFDDF2};Type]

MVC-105489 - [OA] ppSoCCom

[Draft, HW Port;{9C91EC93-E579-DAF4-84C9-608F4873EDC4};Type]

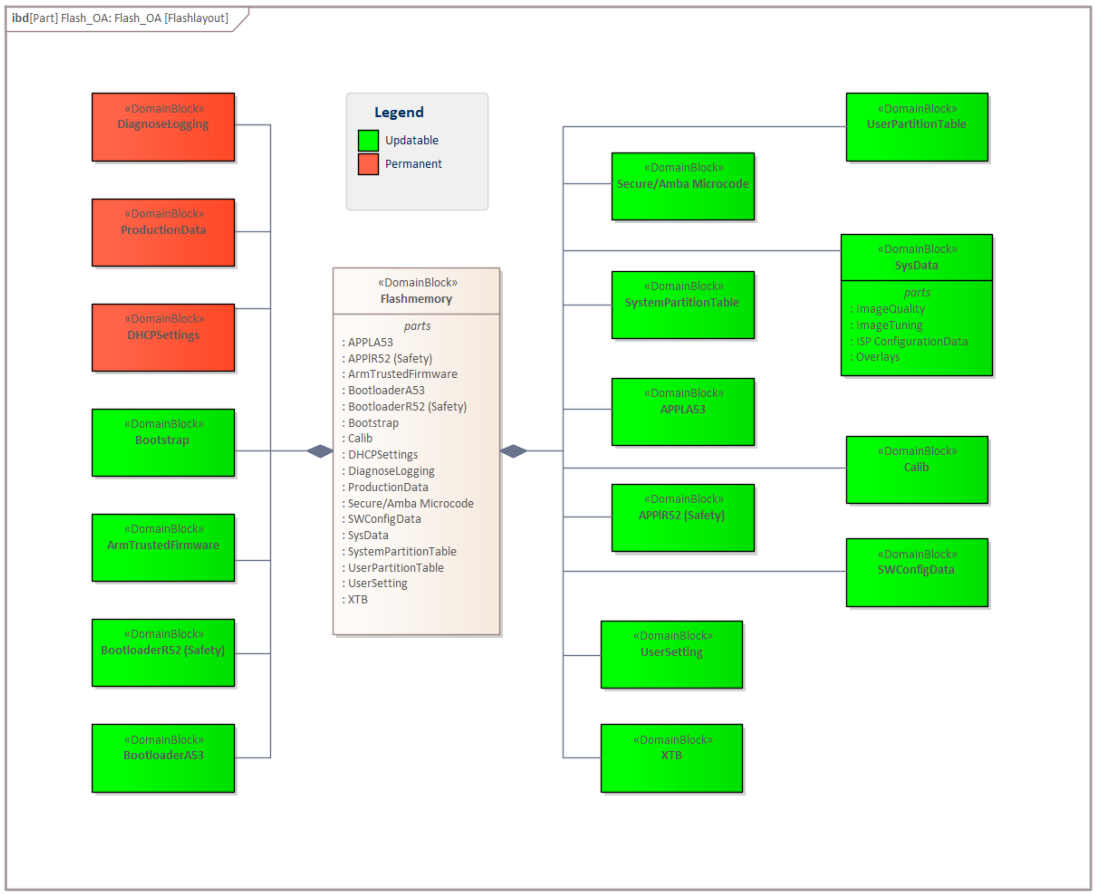
MVC-102850 - ppRGMII HW

[Draft, RGMII;{BEAB1857-0B87-1E57-9D70-A7A304F950CE};Type]

MVC-102783 - Flash_OA

[Specified]

Flashlayout



MVC-103392 - ppPower

[Draft, HW Port;{9C91EC93-E579-DAF4-84C9-608F4873EDC4};Type]

MVC-103379 - flashPower

[Draft, ifPower;{B4E0CCF1-C664-AC45-A778-2FF3A896CAF8};Type]

MVC-103386 - ppMemory

[Draft, HW Port;{9C91EC93-E579-DAF4-84C9-608F4873EDC4};Type]

MVC-103393 - ppDataBus

[Draft, SPI;{A29539D0-FBCE-6D33-B313-4703268EED4B};Type]

MVC-105718 - SecurityData

[Draft, CSProgrammable;{850B3890-48FD-ECC3-AC56-2B6B1BBAD533};Type]

MVC-105720 - RuntimeData

[Draft, [Interface Type]]

MVC-105719 - ProductionData

[Draft, ifMemoryService;{3512C79A-14DF-41b3-96AB-9EFCD06B527C};Type]

MVC-102786 - DRAM_OA

[Specified]

MVC-103388 - ppPower

[Draft, HW Port;{9C91EC93-E579-DAF4-84C9-608F4873EDC4};Type]

MVC-103375 - dramPower

[Draft, ifPower;{B4E0CCF1-C664-AC45-A778-2FF3A896CAF8};Type]

MVC-103387 - ppBuffer

[Draft, HW Port;{9C91EC93-E579-DAF4-84C9-608F4873EDC4};Type]

MVC-102825 - ppDataBus

[Draft, ifMemoryService;{BA0C105B-DA45-DF22-A3EF-1C30669EC9B8};Type]

MVC-102784 - Deserializer_OA

[Specified]

MVC-102881 - ppSoCCamCom

[Draft, HW Port;{9C91EC93-E579-DAF4-84C9-608F4873EDC4};Type]

MVC-102821 - ppI2C

[Draft, I2C;{A5CFB26A-4616-378C-9CDD-009638957548};Type]

MVC-102793 - ppMPFbank

[Draft, HW Port;{9C91EC93-E579-DAF4-84C9-608F4873EDC4};Type]

MVC-102833 - ppMFPHeating

[Draft, ifHeatingSwitch;{D0501131-27AE-45a9-9C83-BA68889955FE};Type]

MVC-102884 - ppCoaxCable

[Draft, HW Port;{9C91EC93-E579-DAF4-84C9-608F4873EDC4};Type]

MVC-102836 - ppDataLinkLayer

[Draft, GMSL;{A98A828F-5535-F52D-A5E7-DCD2CDC7B8F0};Type]

MVC-106228 - ppI2C HW

[Draft, I2C;{A5CFB26A-4616-378C-9CDD-009638957548};Type]

MVC-106229 - ppGPIOforwarding

[Draft, ifGPIOForwarding;{BD398808-97FC-70D8-94CF-CF9A361E3010};Type]

MVC-103378 - desPower

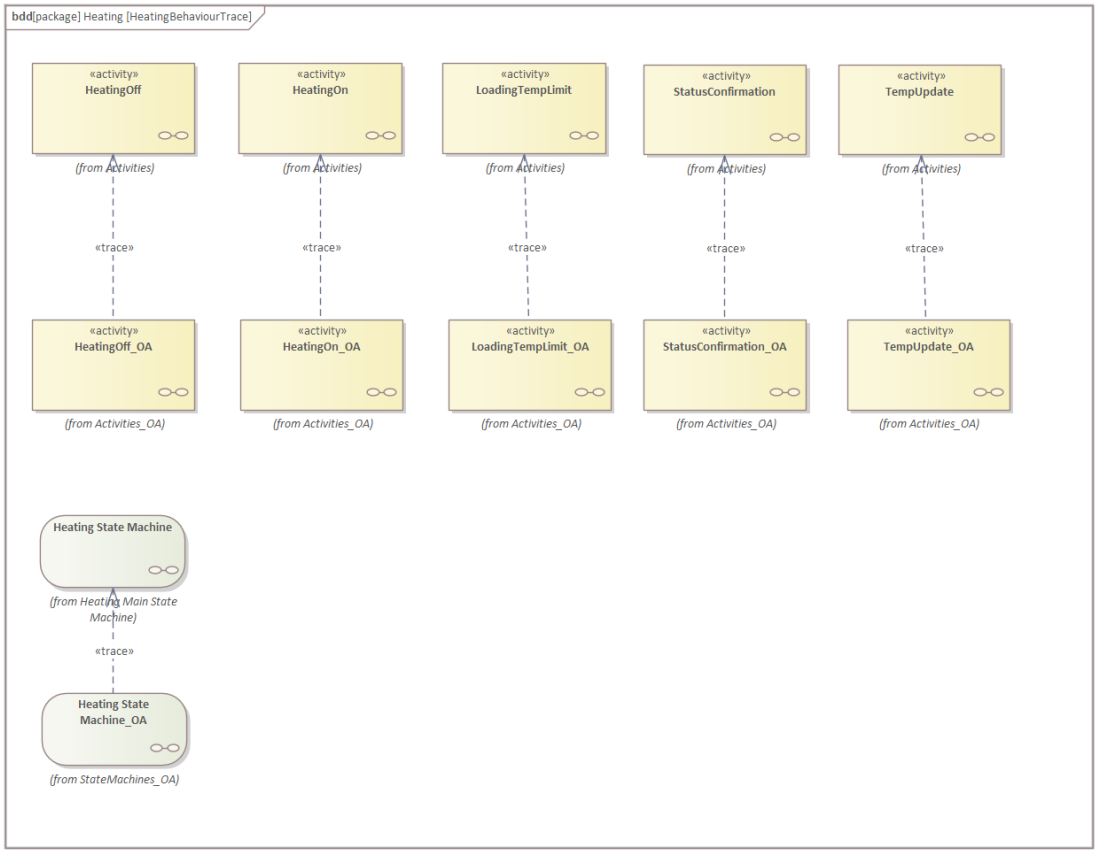
[Draft, ifPower;{B4E0CCF1-C664-AC45-A778-2FF3A896CAF8};Type]

2 Heating

MVC-102861 - Heating

[EA]

HeatingBehaviourTrace



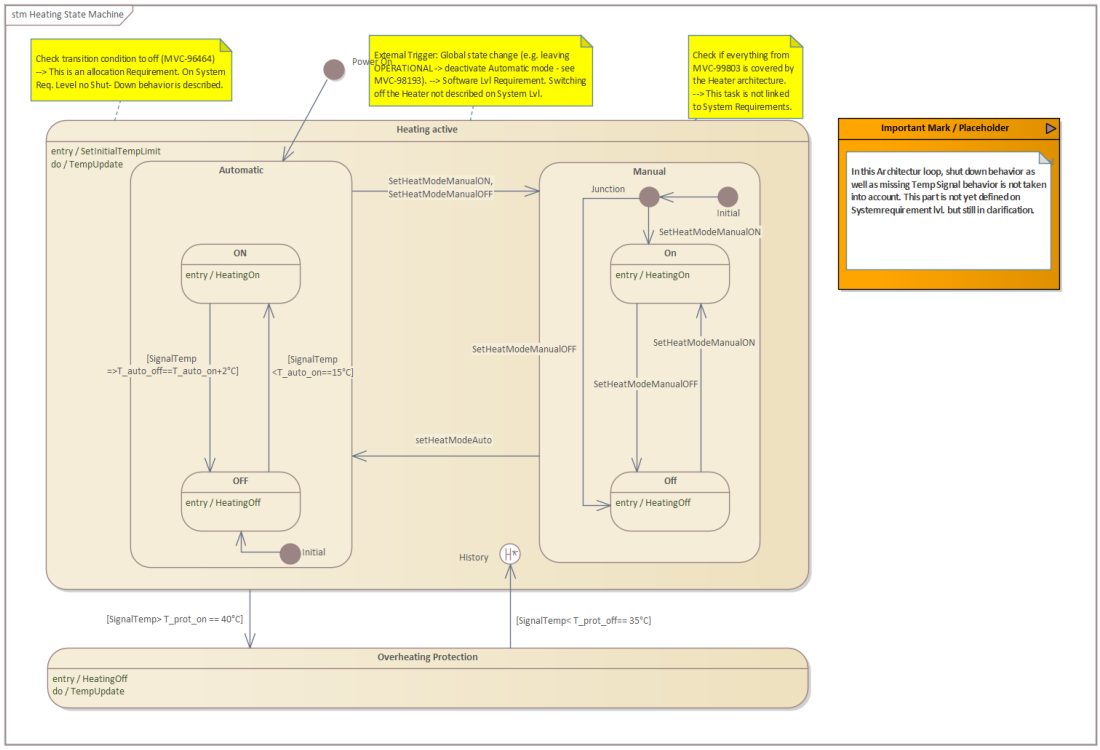
MVC-102867 - StateMachines_OA

[EA **]**

MVC-102878 - Heating State Machine_OA

[Specified **]**

Heating State Machine



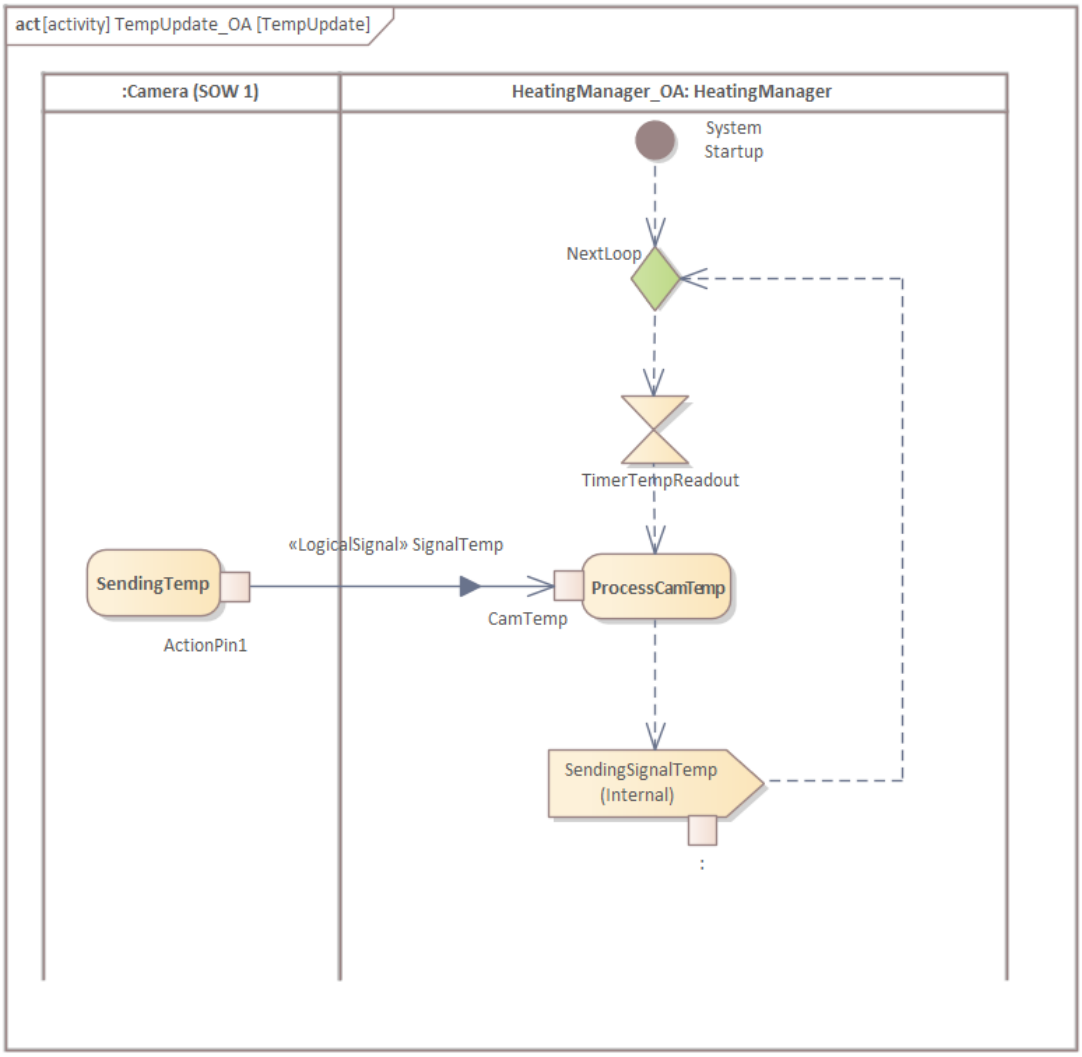
MVC-102865 - Activities_OA

[EA]

MVC-102876 - TempUpdate_OA

[Specified]

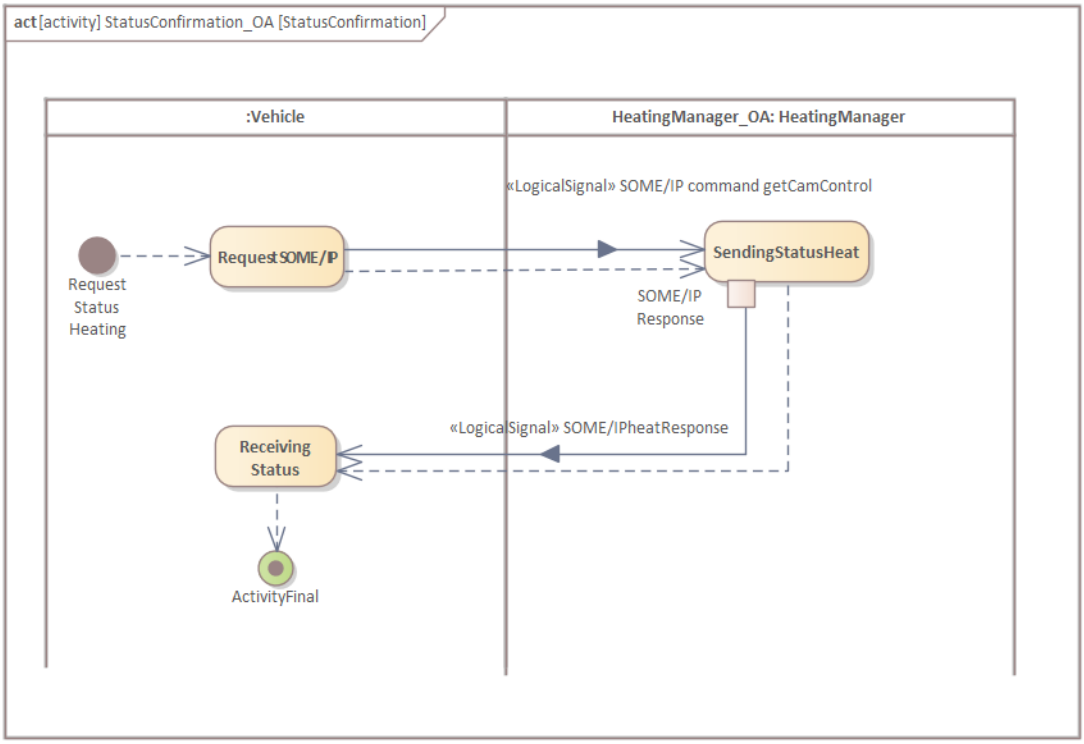
TempUpdate



MVC-102877 - StatusConfirmation_OA

[Specified **]**

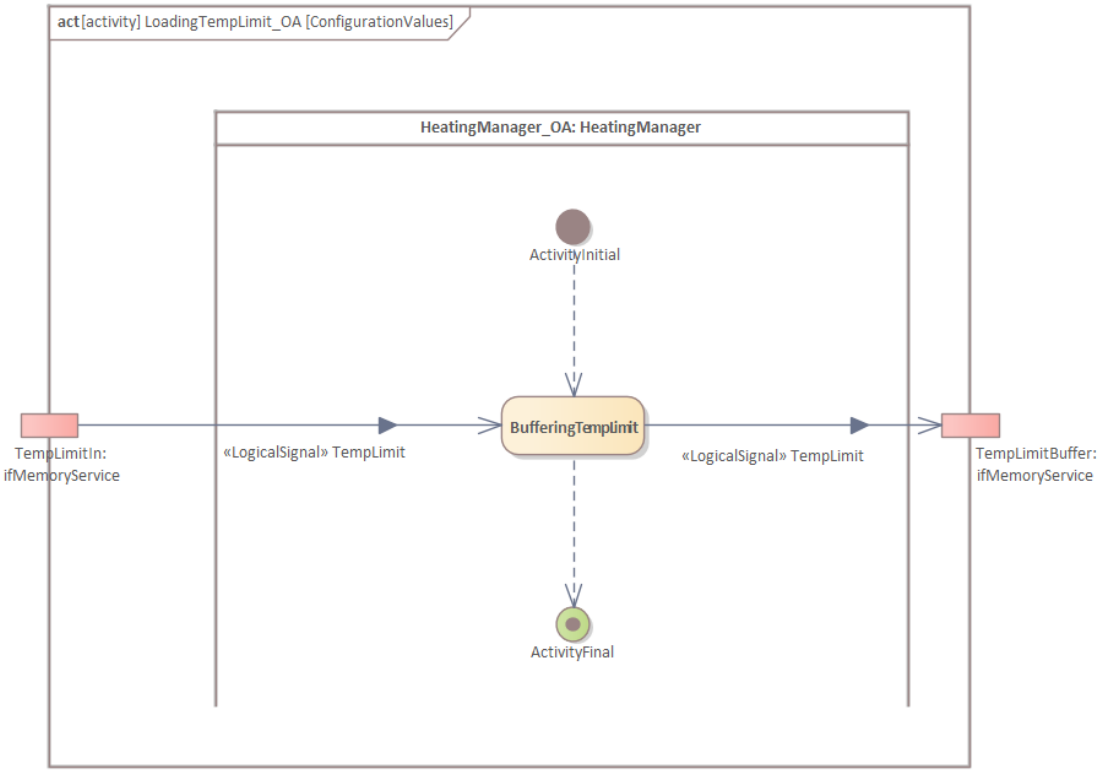
StatusConfirmation



MVC-102871 - LoadingTempLimit_OA

[Specified]

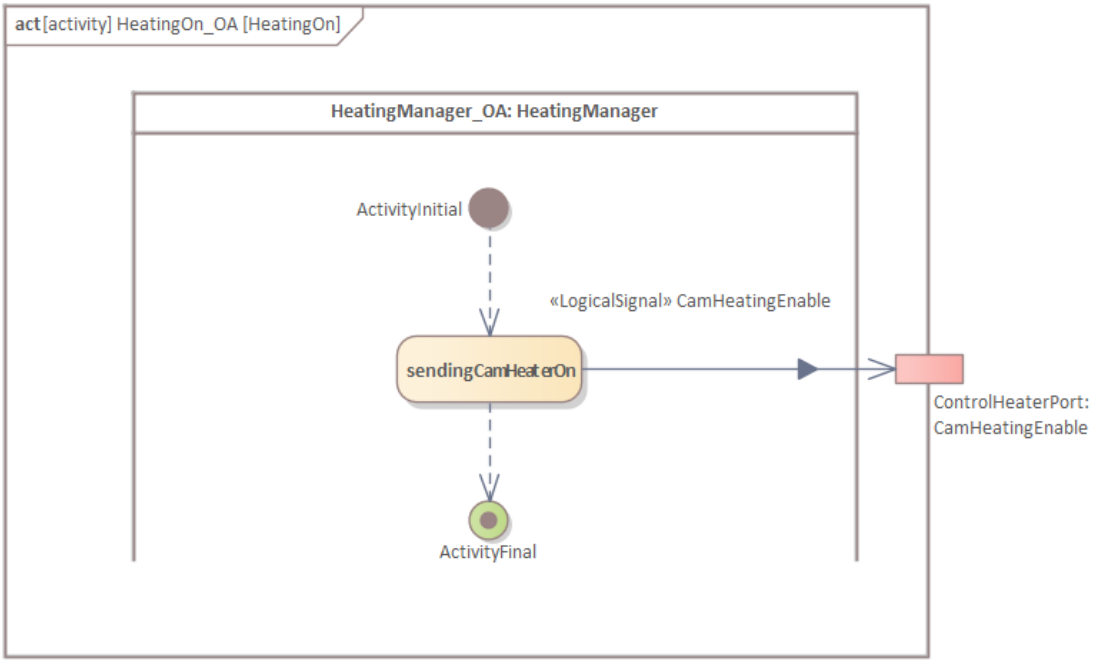
ConfigurationValues



MVC-102874 - HeatingOn_OA

[Specified]

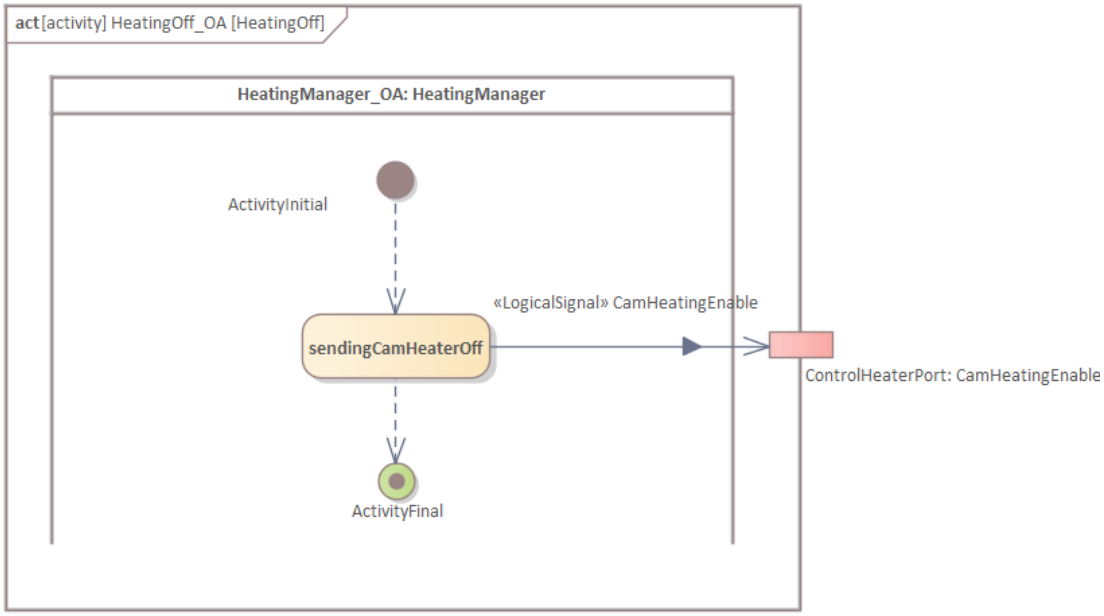
HeatingOn



MVC-102873 - HeatingOff_OA

[Specified]

HeatingOff

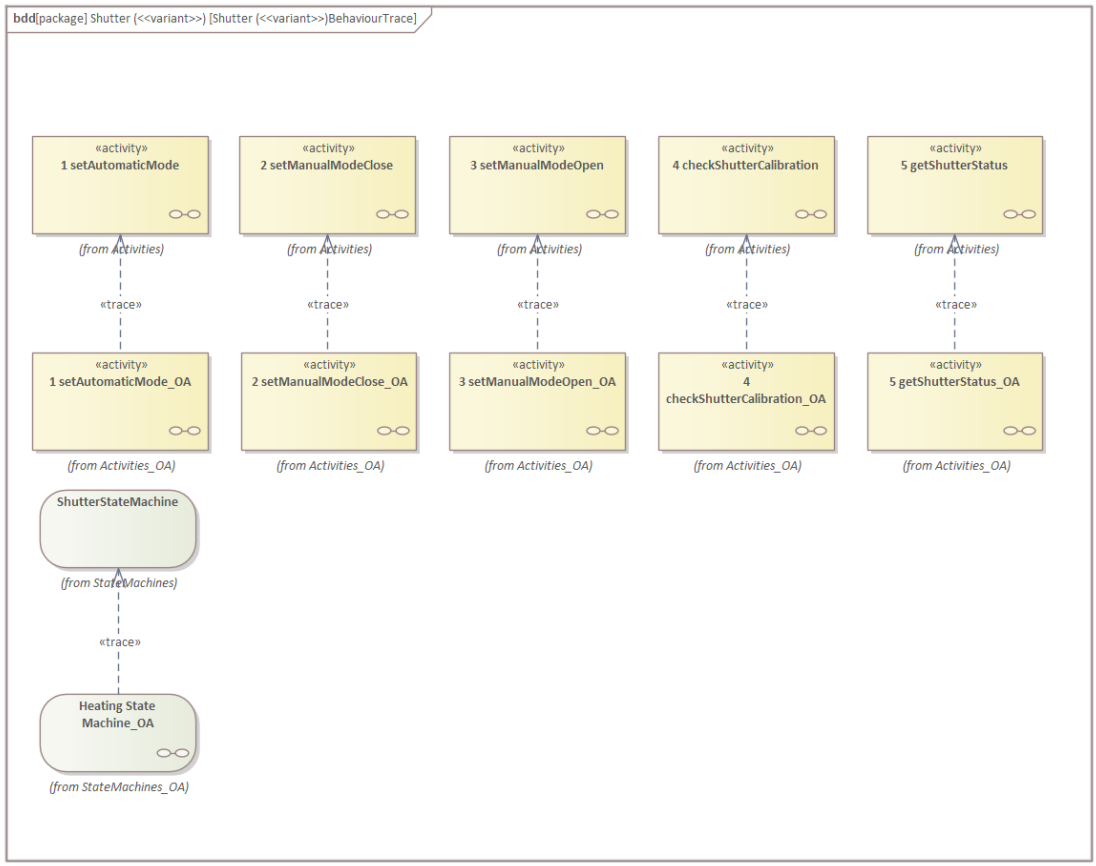


3 Shutter

MVC-102834 - Shutter (<<variant>>)

[EA]

Shutter (<<variant>>)BehaviourTrace



MVC-102875 - UseCases_OA

[EA]

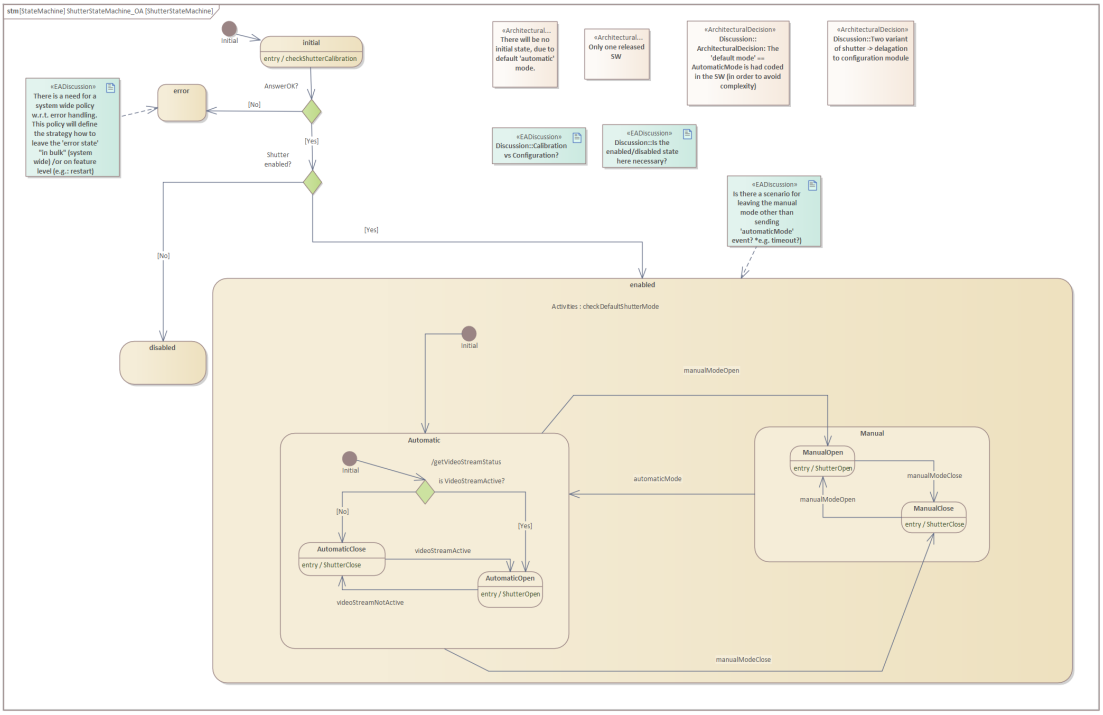
MVC-102817 - StateMachines_OA

[EA]

MVC-102846 - ShutterStateMachine_OA

[Specified]

ShutterStateMachine



MVC-102852 - Sequences_OA

[EA]

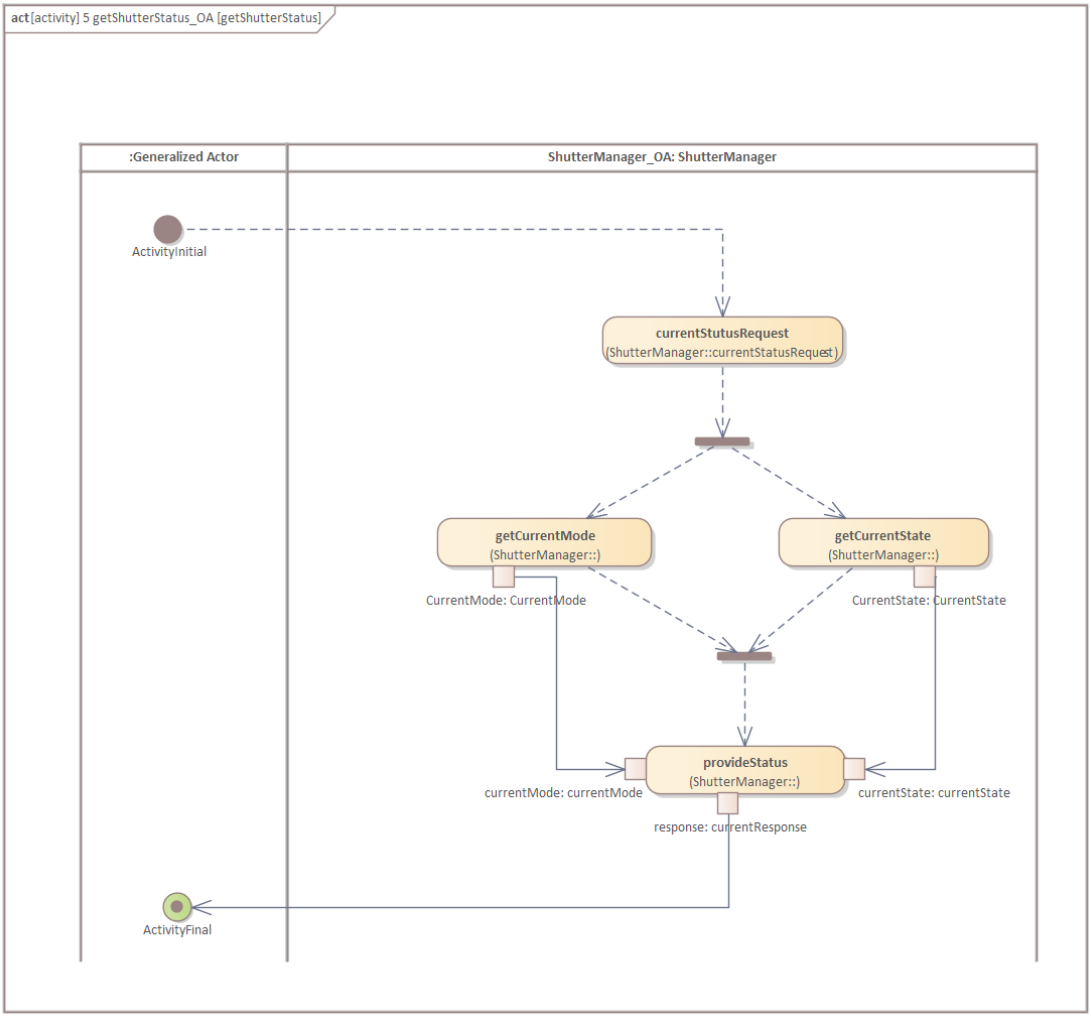
MVC-102820 - Activities_OA

[EA]

MVC-102897 - 5 getShutterStatus_OA

[Specified]

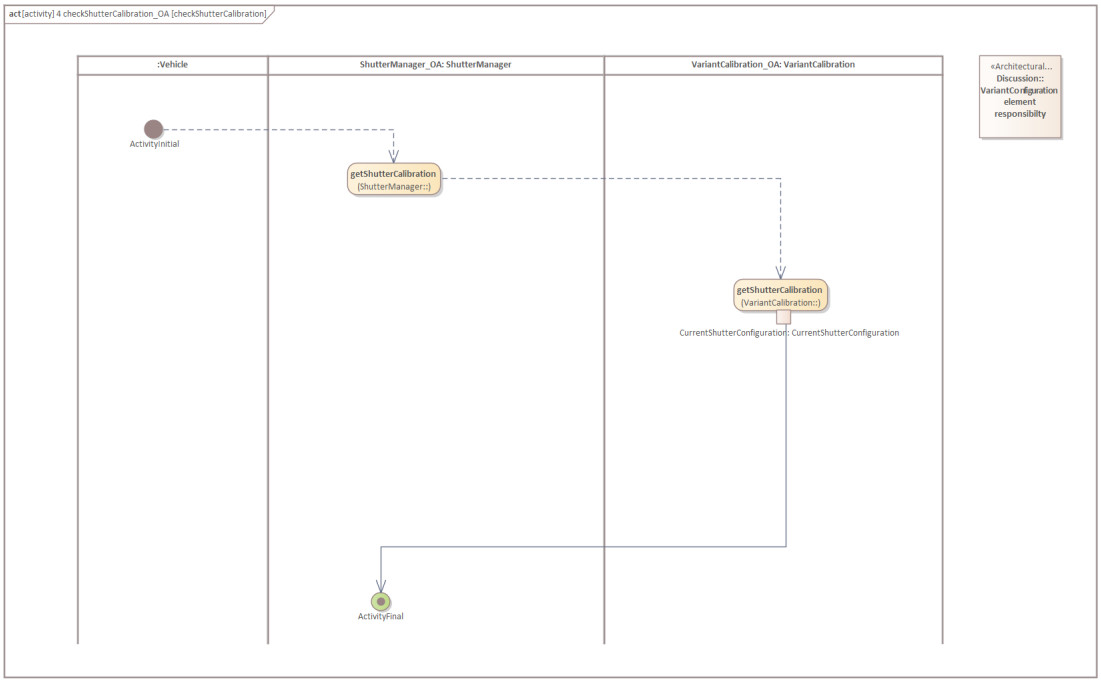
getShutterStatus



MVC-102851 - 4 checkShutterCalibration_OA

[Specified **]**

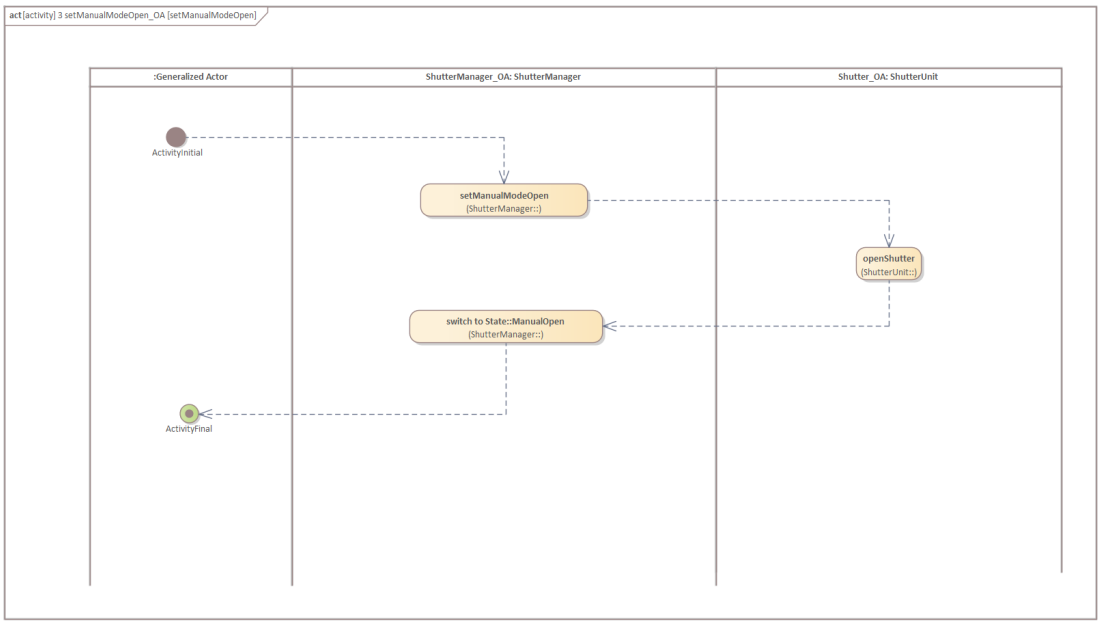
checkShutterCalibration



MVC-102853 - 3 setManualModeOpen_OA

[Specified **]**

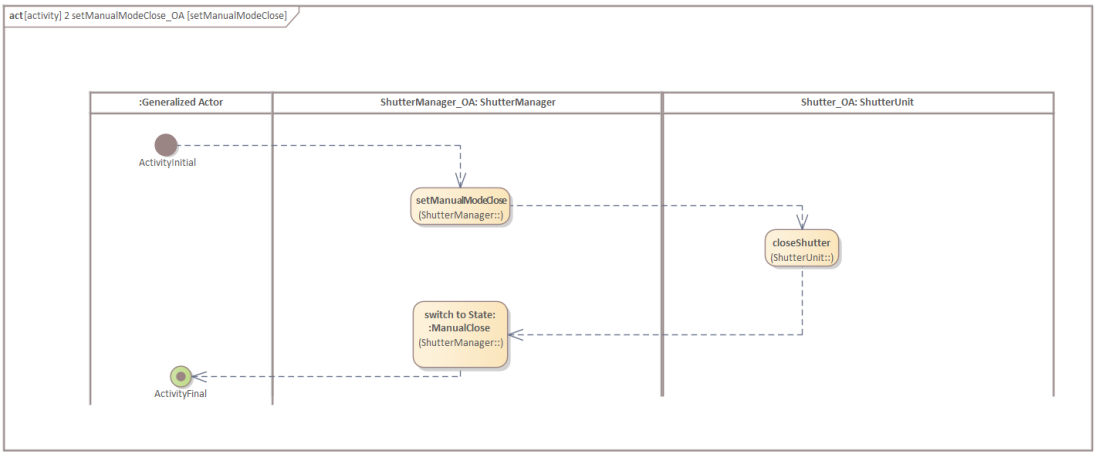
setManualModeOpen



MVC-102898 - 2 setManualModeClose_OA

[Specified **]**

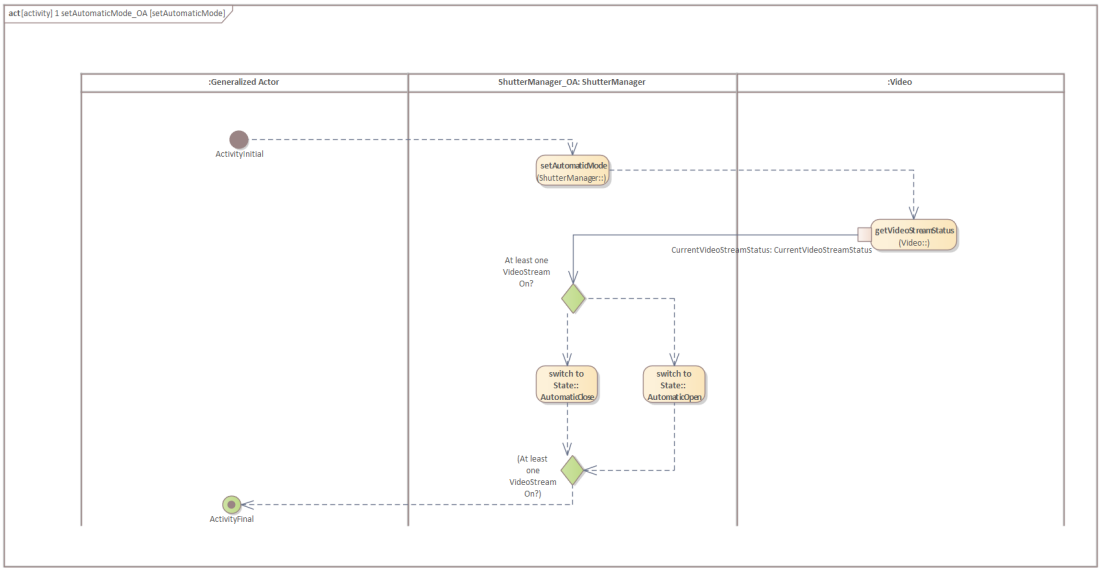
setManualModeClose



MVC-102857 - 1 setAutomaticMode_OA

[Specified]

setAutomaticMode

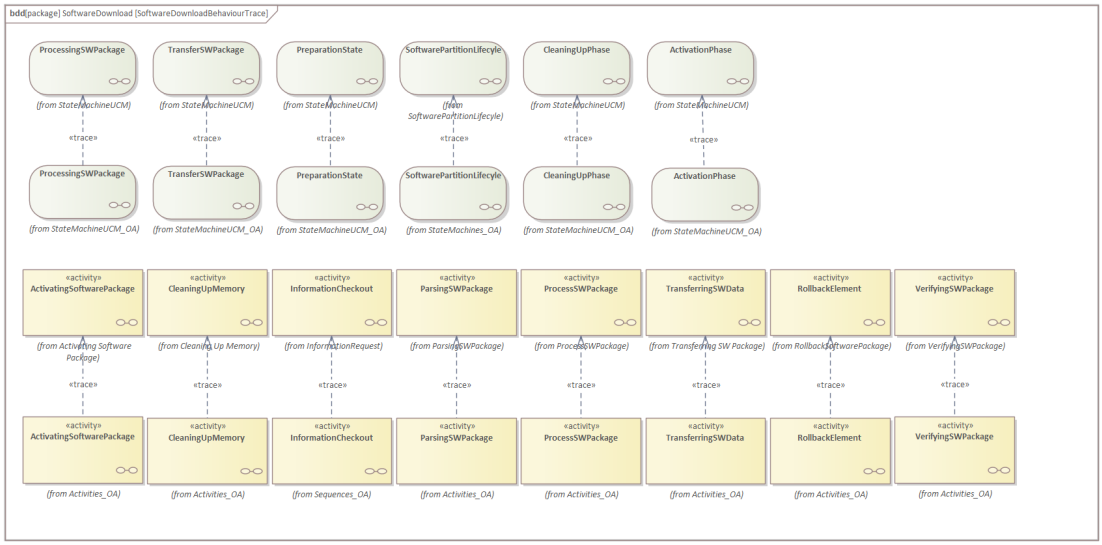


4 Software Download

MVC-105742 - SoftwareDownload

[EA]

SoftwareDownloadBehaviourTrace



MVC-105744 - StateMachines_OA

[EA]

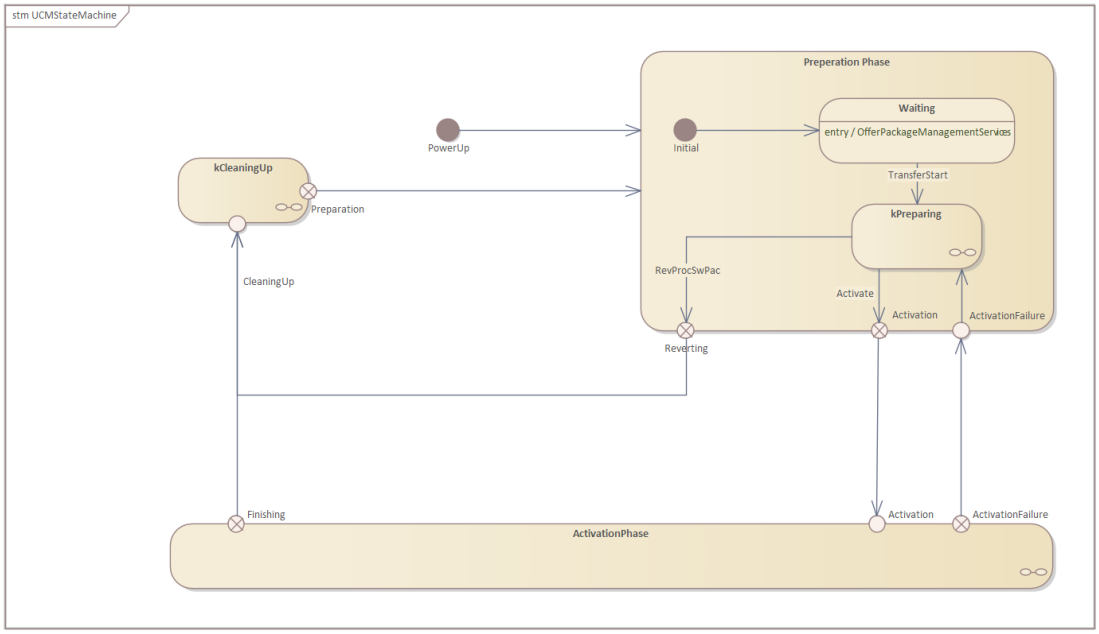
MVC-105747 - StateMachineUCM_OA

[EA]

MVC-105758 - UCMStateMachine_OA

[Draft]

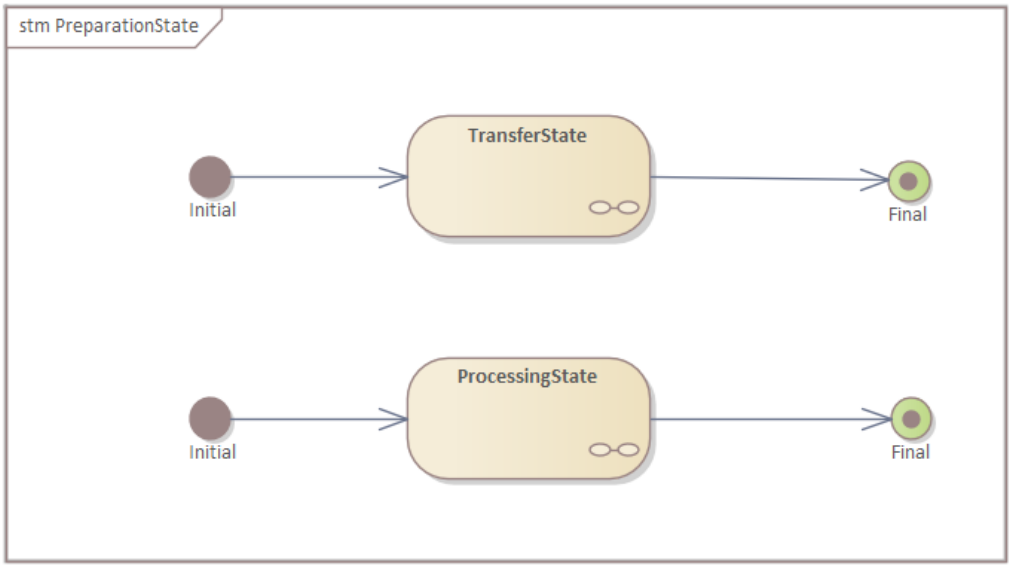
UCMStateMachine



MVC-105736 - PreparationState

[Draft]

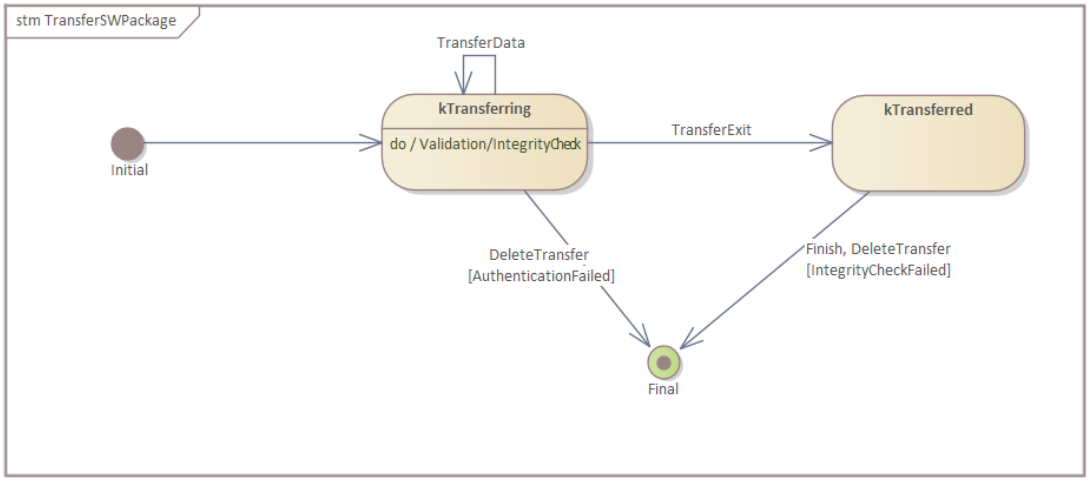
PreparationState



MVC-105738 - TransferSWPackage

[Draft]

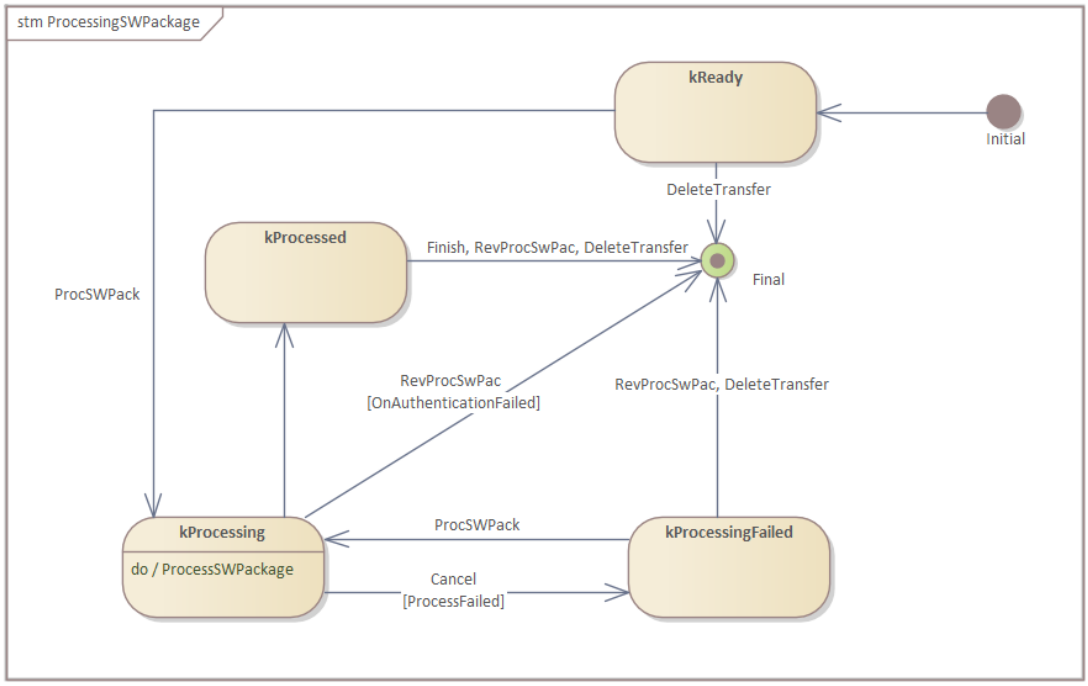
TransferSWPackage



MVC-105739 - ProcessingSWPackage

[Draft]

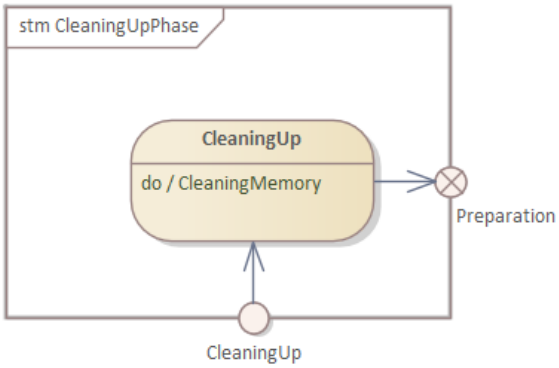
ProcessingSWPackage



MVC-105737 - CleaningUpPhase

[Draft]

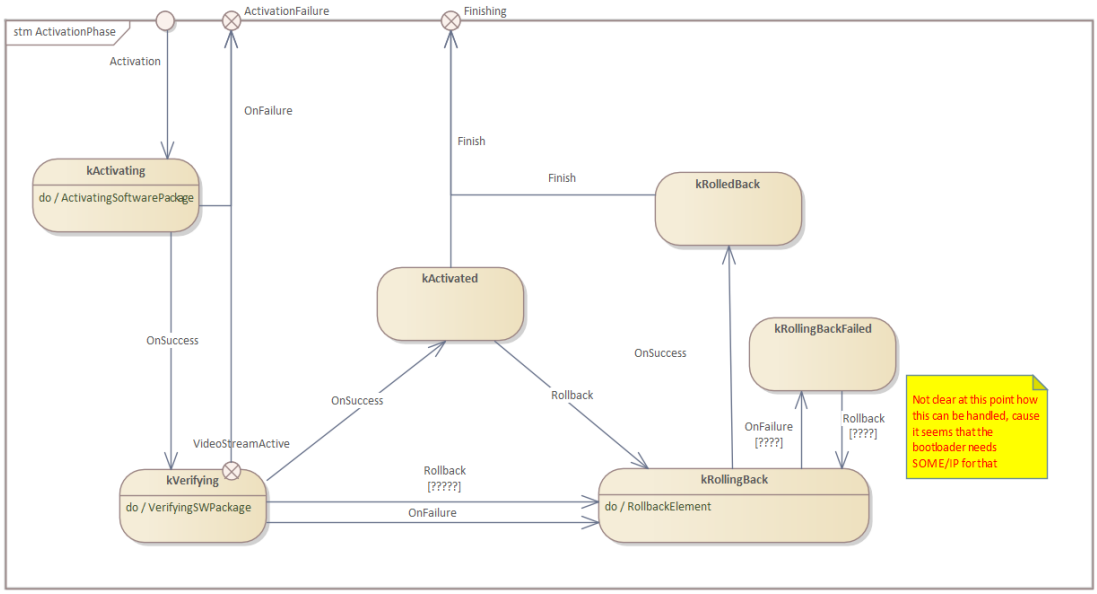
CleaningUpPhase



MVC-105735 - ActivationPhase

[Draft]

ActivationPhase



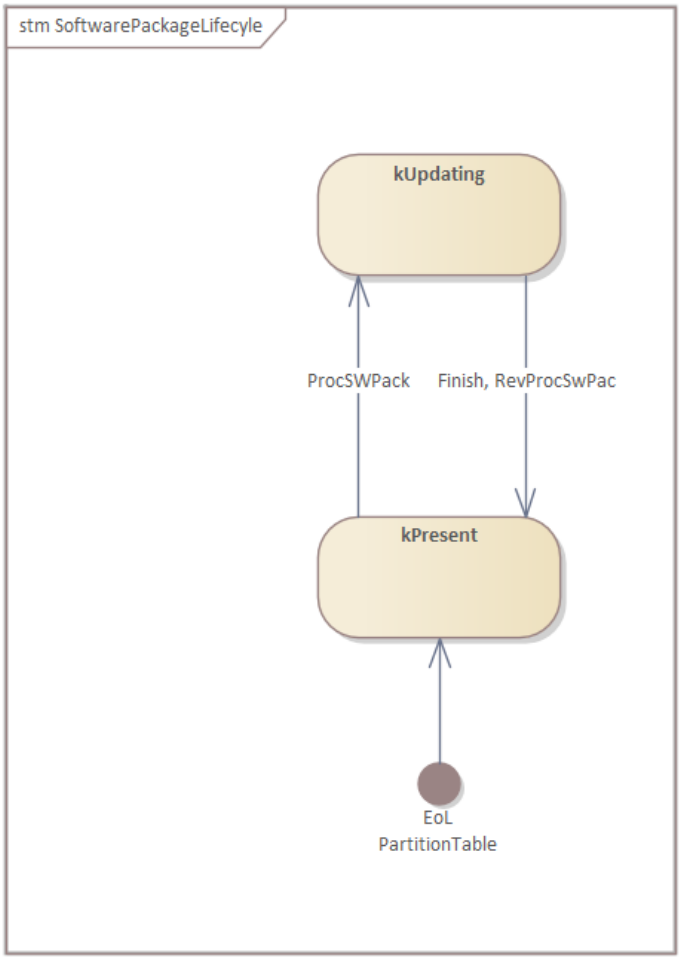
MVC-105757 - Triggers

[EA]

MVC-105748 - SoftwarePartitionLifecyle

[Draft]

SoftwarePackageLifecyle



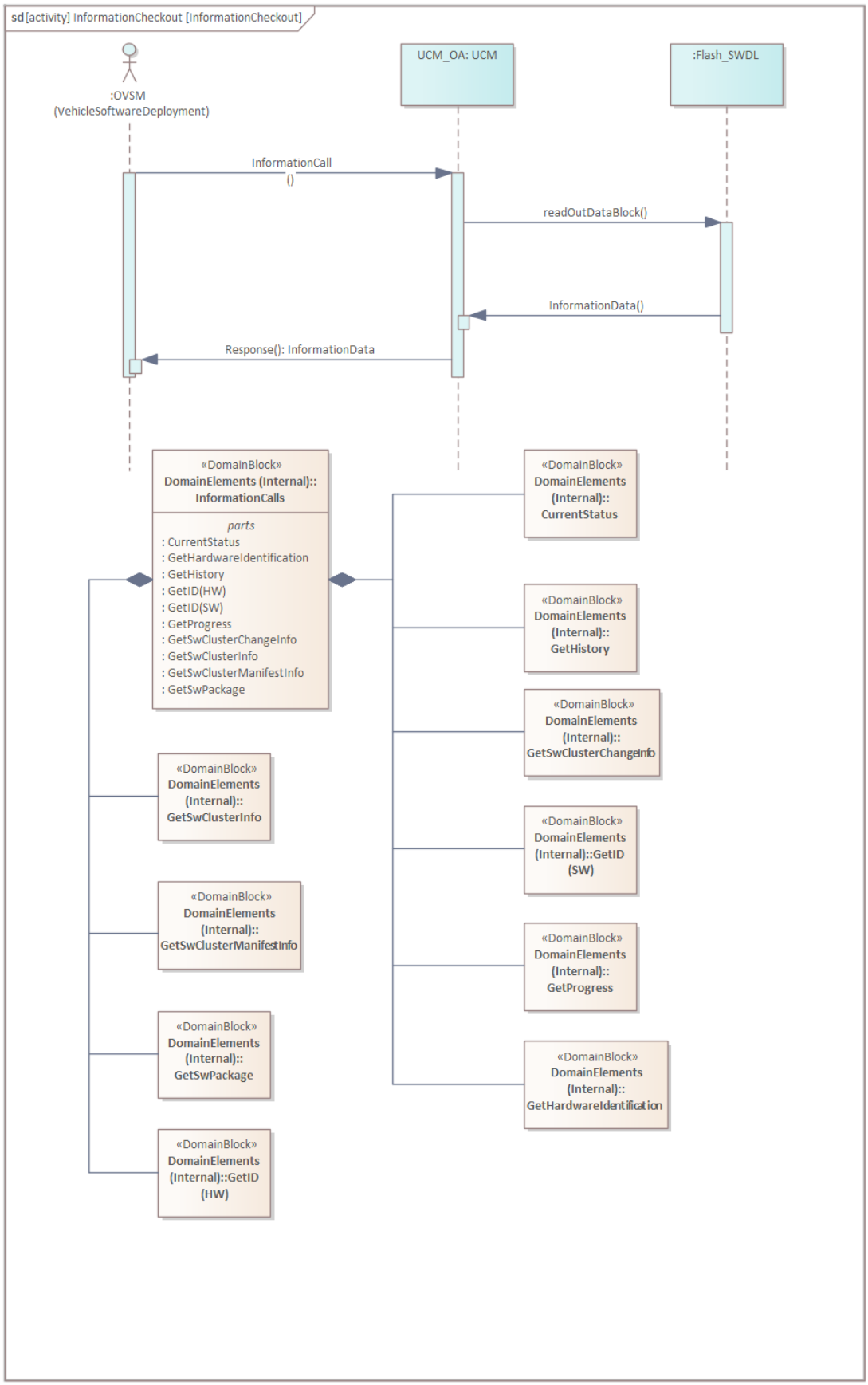
MVC-105743 - Sequences_OA

[EA **]**

MVC-105746 - InformationCheckout

[Draft **]**

InformationCheckout



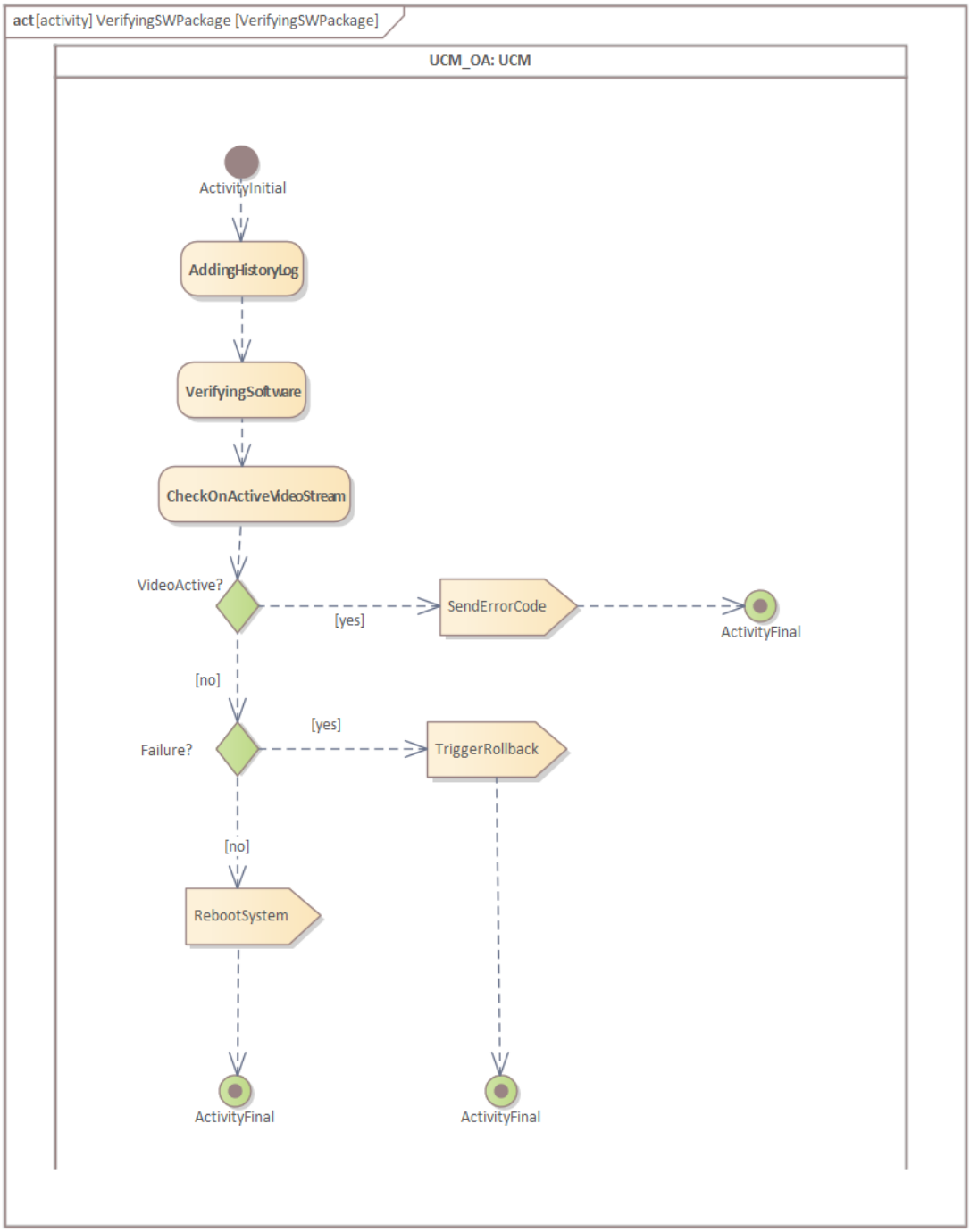
MVC-105745 - Activities_OA

[EA]

MVC-105756 - VerifyingSWPackage

[Draft]

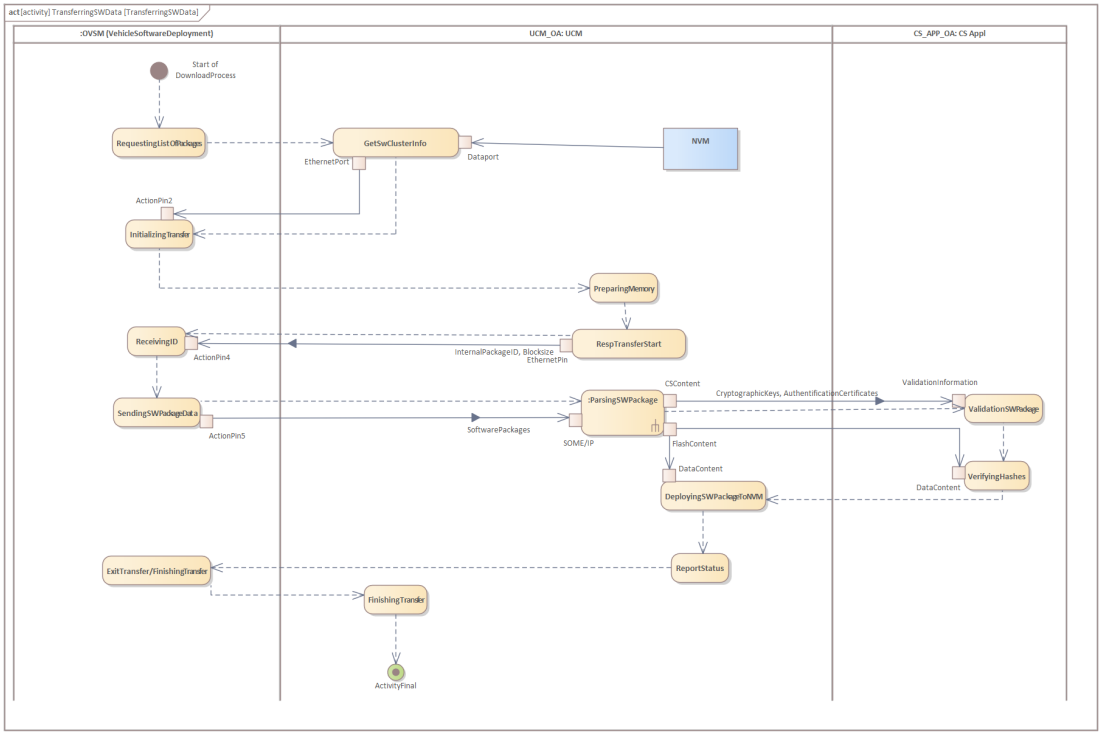
VerifyingSWPackage



MVC-105750 - TransferringSWData

[Draft]

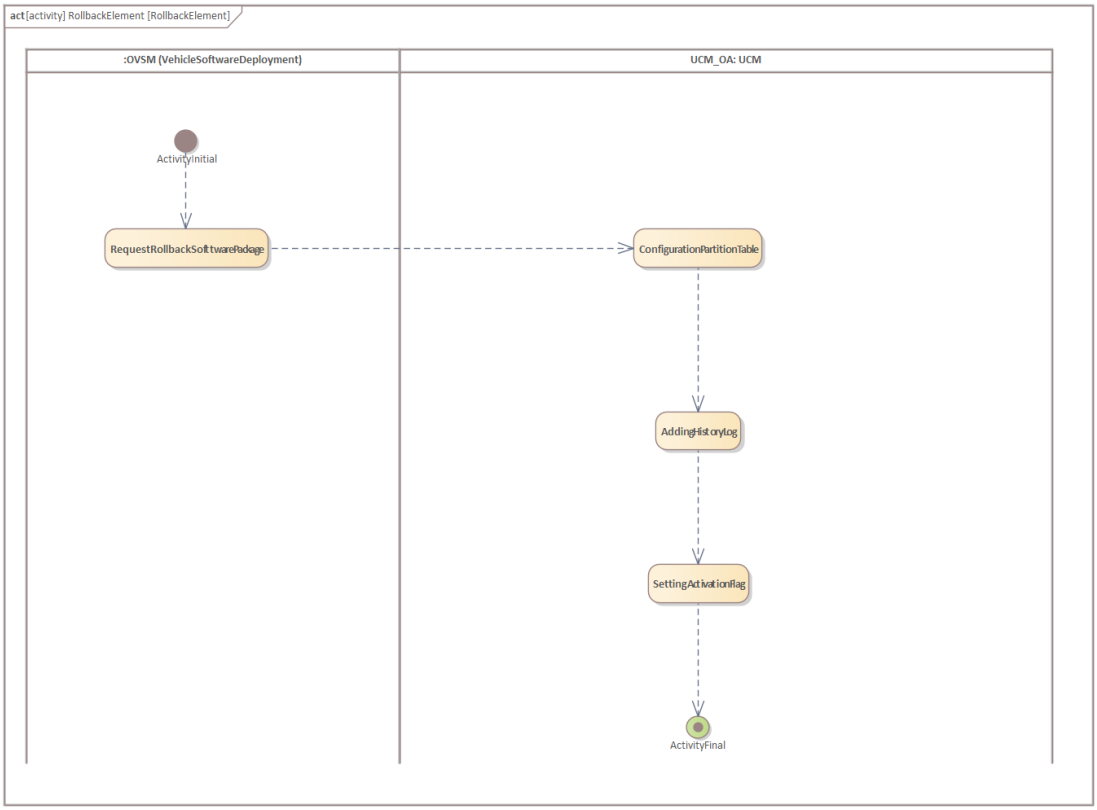
TransferringSWData



MVC-105753 - RollbackElement

[Draft]

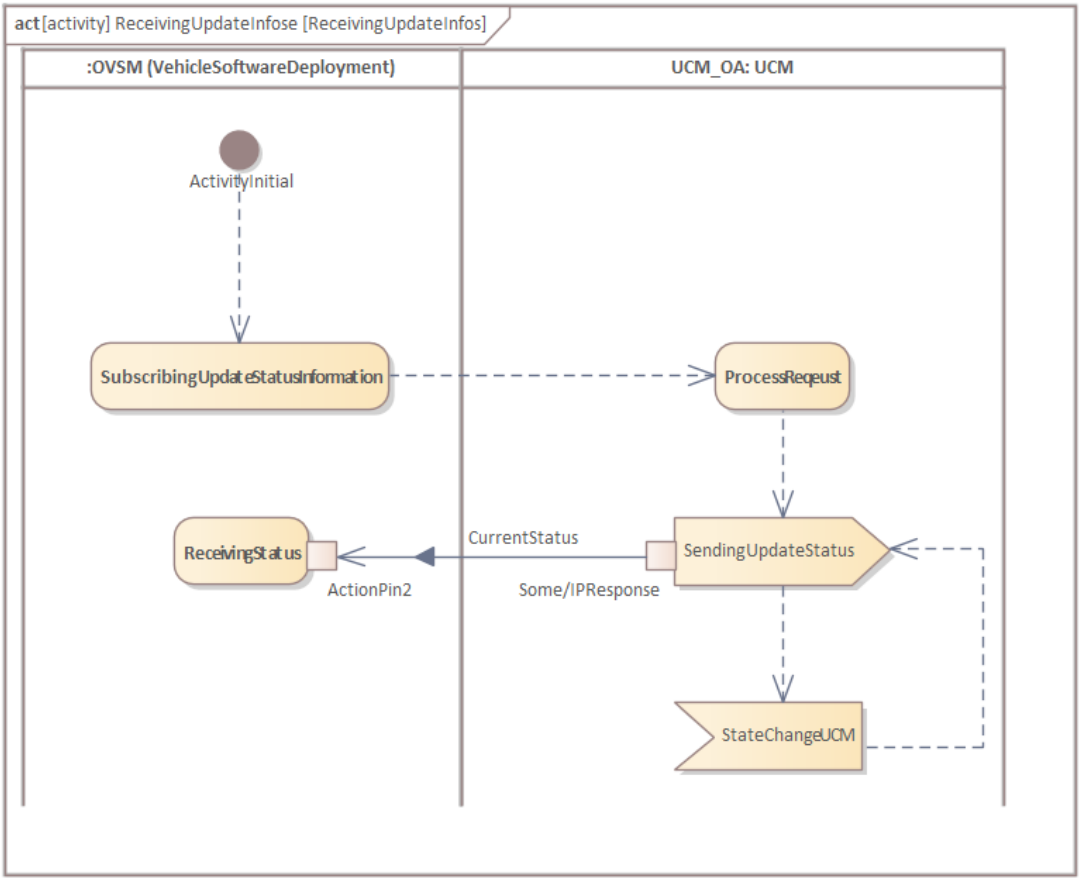
RollbackElement



MVC-105749 - ReceivingUpdateInfose

[Draft]

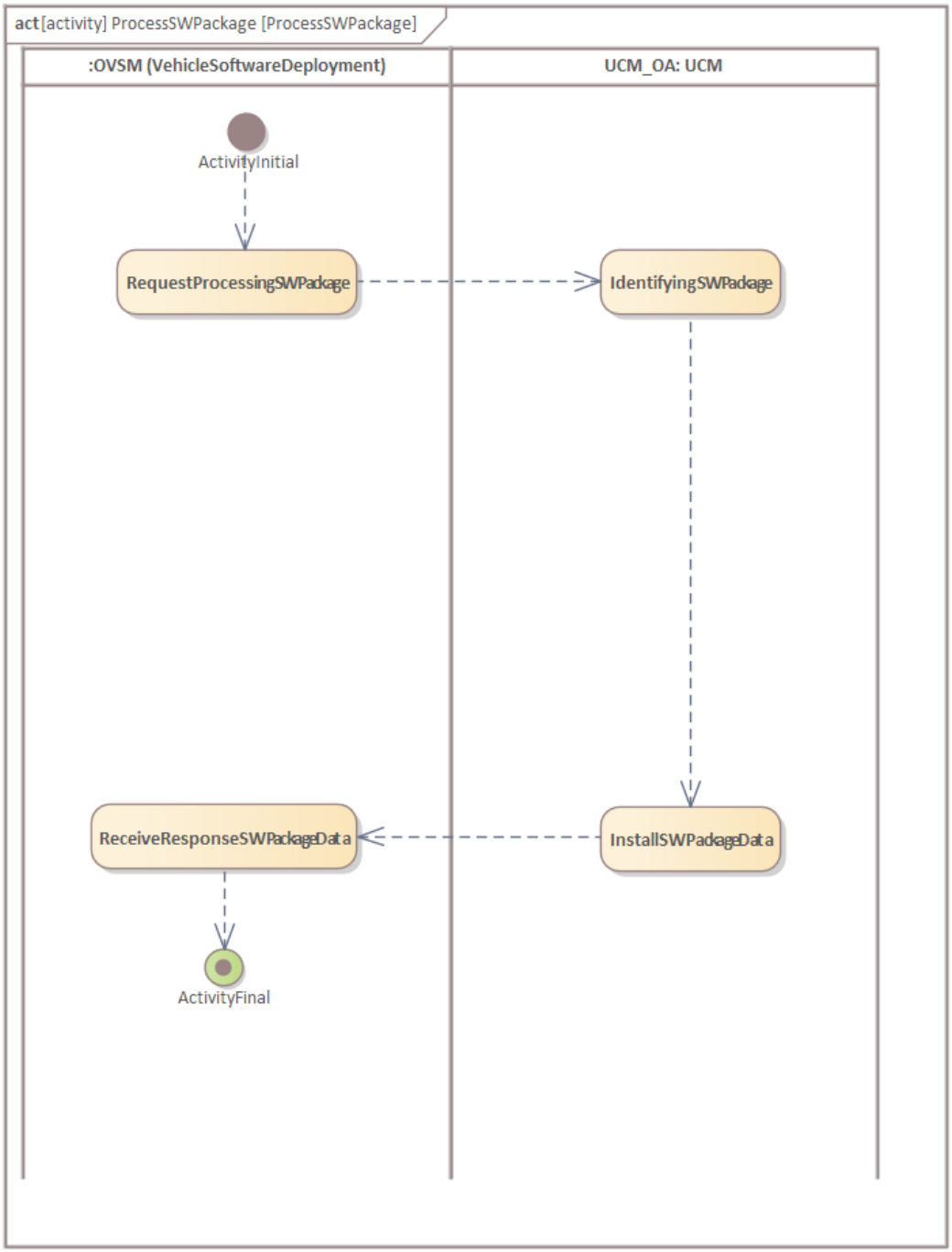
ReceivingUpdateInfos



MVC-105755 - ProcessSWPackage

[Draft]

ProcessSWPackage



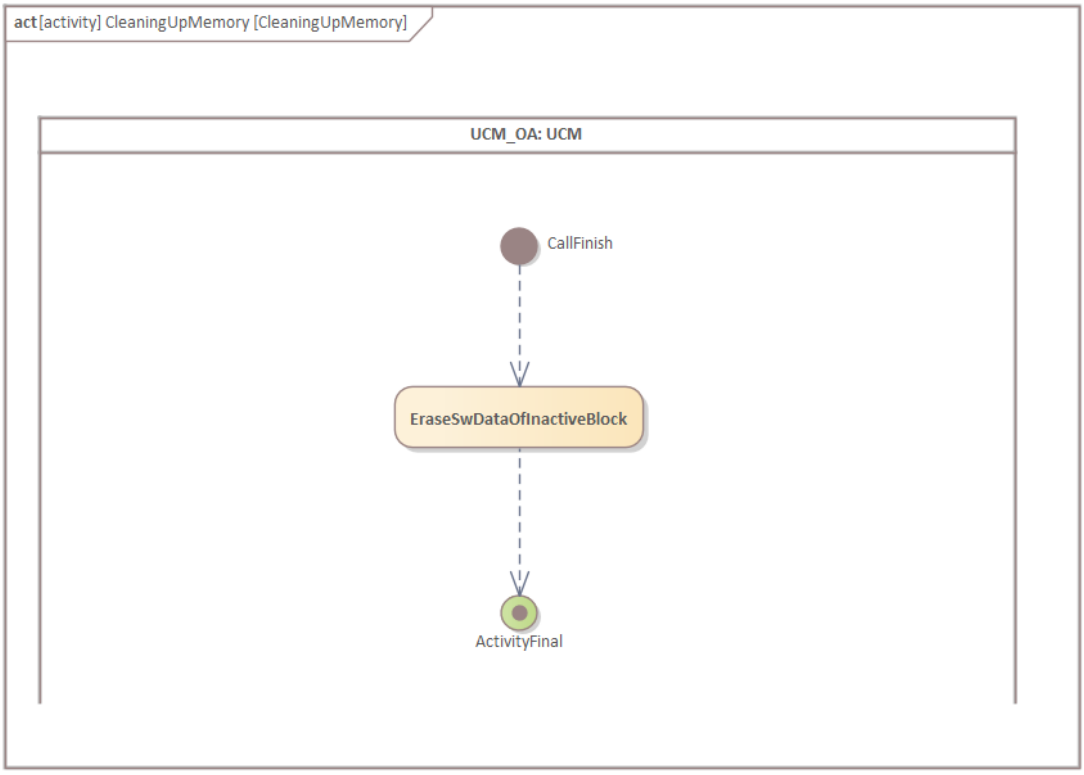
MVC-105752 - ParsingSWPackage

[Draft]

MVC-105754 - CleaningUpMemory

[Draft]

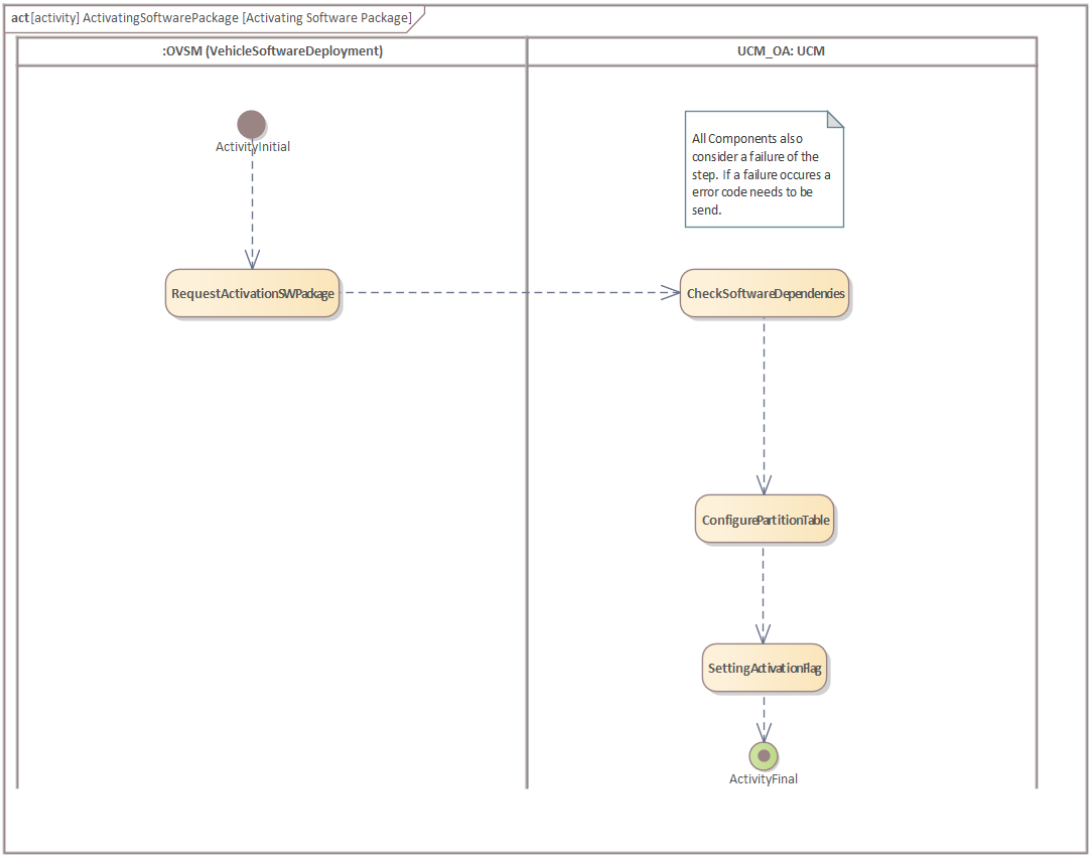
CleaningUpMemory



MVC-105751 - ActivatingSoftwarePackage

[Draft]

Activating Software Package


act [activity] ActivatingSoftwarePackage [Activating Software Package]
:OVSM (VehicleSoftwareDeployment)
UCM_OA: UCM
ActivityInitial
All Components also consider a failure of the step. If a failure occures a error code needs to be send.
RequestActivationSWPackage
CheckSoftwareDependencies
ConfigurePartitionTable
SettingActivationFlag
ActivityFinal